

nVidia's nForce 3 250 GB chipset, and one each on VIA's K8T800 and the SiS 760GX. Let's begin with features first, and then turn our attention to performance.

Features

All these boards support AMD 754-pin processors up to the maximum speed available. On the memory front, since the memory controller is integrated into the processor core, it determines the speed of the memory; in this case, it is DDR 400 MHz.

In other words, all these boards support DDR 400 MHz memory in single-channel mode only. Here, the HyperTransport bus is capped at 800 MHz, which otherwise runs at 1000 MHz for 939-pin processors. Except for the SiS chipset, all boards support 3 GB of memory; the Asus K8S-MX can support a maximum of 2 GB.

All the boards have two IDE connectors that can connect four IDE devices. Asus' K8V-SE Deluxe has gone one step further, offering one extra IDE connector for RAID. MSI's K8N Neo Platinum and Asus' K8V-SE have four SATA ports, whereas the K8N-E Deluxe can support up to six SATA drives in RAID 0, 1, 0+1 and JBOD functionalities. Gigabyte's GA-K8NS and the Asus K8S-MX support two SATA drives, and are limited to RAID 0 and 1 capabilities.

Except for the ASUS K8S-MX, which has an integrated video chip based on the SiS 760GX chipset, no other board has an onboard video controller.

The Asus K8N-E and K8V-SE, and MSI's K8N Neo Platinum motherboards have FireWire ports. Eight USB 2.0 ports are supported by almost all the motherboards; however, only four to six are on the back panel, and the rest are available via headers on the board.

On the Intel side, we have seen that the Gigabit network adapter is present on all 915-chipset-based motherboards; however, on the AMD platform, only the Deluxe or Platinum series boards have them, while the rest use the older 10/100 standard. If you transfer huge files over the network, or if you are a movie-editing house or an E-learning developer, ensure that Gigabit Ethernet is supported by your motherboard.

Six-channel audio is present on almost all motherboards, and these were no exception. We did not compare the fidelity offered by either these boards or by the Intel boards that support High-Definition Audio, and hence cannot comment on the sound quality.

Five PCI slots are common on these boards, but the Asus K8S-MX had only two since it is based on a micro-ATX form factor.

MSI's K8N Neo Platinum sported the best component layout. In fact, the layout was completely different from the norm. The processor area was free of all intruding capacitors, and the storage ports were placed perfectly for hindrance-free operation. The component layout on the Asus K8V-SE Deluxe is also nice—a similar approach is visible on Gigabyte's GA-K8NS. Due to its small form factor, the layout on the Asus K8S-MX was a little cramped.

The Deluxe and Platinum tags ought to tell you that the package is packed with features. And this was true, as Asus' K8N-E Deluxe, MSI's K8N Neo Platinum and Asus' K8V-SE Deluxe were generous on their packaging and features.

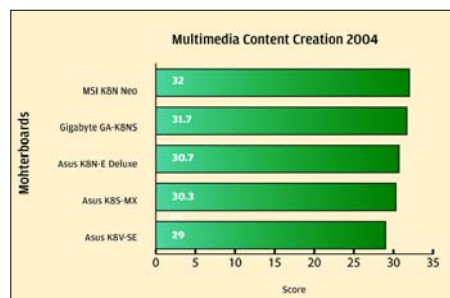
Performance

Having taken a look at the feature set these motherboards offer, let's

move on to the benchmarks and see how the chipsets performed.

Multimedia Content Creation 2004

MSI's K8N Neo Platinum topped in this system-level benchmark, indicating that this board has what it takes to be the best and does not bog down when all sub-systems are stressed. Gigabyte's GA-K8NS is off-target by a very small margin from the leader, MSI. The remaining Asus boards put up a relatively decent showing, with the dif-



ference between the Asus K8V-SE and the MSI K8N Neo not being more than 10 per cent.

PCMark 2004 CPU Sub-system

In the CPU test, the ASUS K8N-E Deluxe emerged on top, displacing the MSI K8N Neo Platinum to the second position—albeit only by a small margin of 0.5 per cent. The race for third place was close: Gigabyte's GA-K8NS managed to edge ahead of the Asus K8S-MX by an extremely small margin.

In the last slot was the Asus K8V-SE deluxe. The point to note here is that the top three were based on the nForce 3 250 GB, and it seems that this nVidia chipset has a good CPU sub-system. Between the VIA and SiS chipsets, it's hard to give a clear verdict since we had only one board per chipset.

PCMark 2004 Memory Sub-system

The results were pretty much the same again: Asus' K8N-E Deluxe took the top slot, to be closely followed by MSI's K8N Neo Platinum. The VIA chipset-powered Asus K8V-SE beat Gigabyte's GA-K8NS by a convincing margin, whereas the SiS-based Asus K8S-MX came in last.

For the memory sub-system, a lot depends up on the actual layout of the memory bus on the

Scoreboard

Motherboard	Basic Specifications	Score (100)	Price *
AMD Platform (Socket 754 Motherboards)			
Asus K8N-E Deluxe	nForce3, 6 SATA W/T RAID, Gigabit, 7.1 Audio	77.97	8736
Asus K8V-SE Deluxe	VIA K8T800, 4 SATA W/T RAID, Gigabit, 5.1 Audio	76.42	8320
Asus K8S-MX	SiS 760GX, 2 SATA W/T RAID, 10/100, 5.1 Audio, Onboard Video	77.56	4992
Gigabyte GA-K8NS	nForce3, 2 SATA W/T RAID, 10/100, 5.1 Audio	77.65	5800
MSI K8N Neo Platinum	nForce3, 4 SATA W/T RAID, Gigabit, 7.1 Audio	78.67	8000
Intel Platform (i915 Motherboards)			
Asus P5GD1-VM	DDR 400, Onboard Video, 4 SATA, X16 PCI-E Lane, 7.1 Audio	68.98	7904
Gigabyte GA-8I915G Duo	DDR and DDR-II 533/400, onboard video, 4 SATA, X16 PCI-E, 7.1 Audio	78.77	6000
Gigabyte GA-8I915G-MF	DDR 400, Onboard Video, 4 SATA, X16 PCI-E Lane, 7.1 Audio	76.61	5300
MSI 915G Neo 2 Platinum	DDR-II 533/400, onboard video, 4 SATA W/T RAID, X16 PCI-E, 7.1 Audio	74.71	9700
MSI 915G Combo	DDR and DDR-II 533/400, onboard video, 4 SATA, X16 PCI-E, 7.1 Audio	76.19	6500
MSI 915P Neo 2 Platinum	DDR-II 533/400, 4 SATA W/T RAID, X16 PCI-E, 7.1 Audio	73.31	9600

For the entire scoreboard, please visit www.thinkdigit.com